Chapter 14.3: Databases

BMM 105 Introduction to Computer Engineering

Slides From: Invitation to Computer Science, Fourth Edition

Databases

- Bit
  - Most basic unit of data
  - Combined into groups of eight called bytes
- Fields
  - Group of bytes
- Record
  - Collection of related fields

Databases (continued)

- Data file
  - Stores related records
- Database
  - Made up of related files

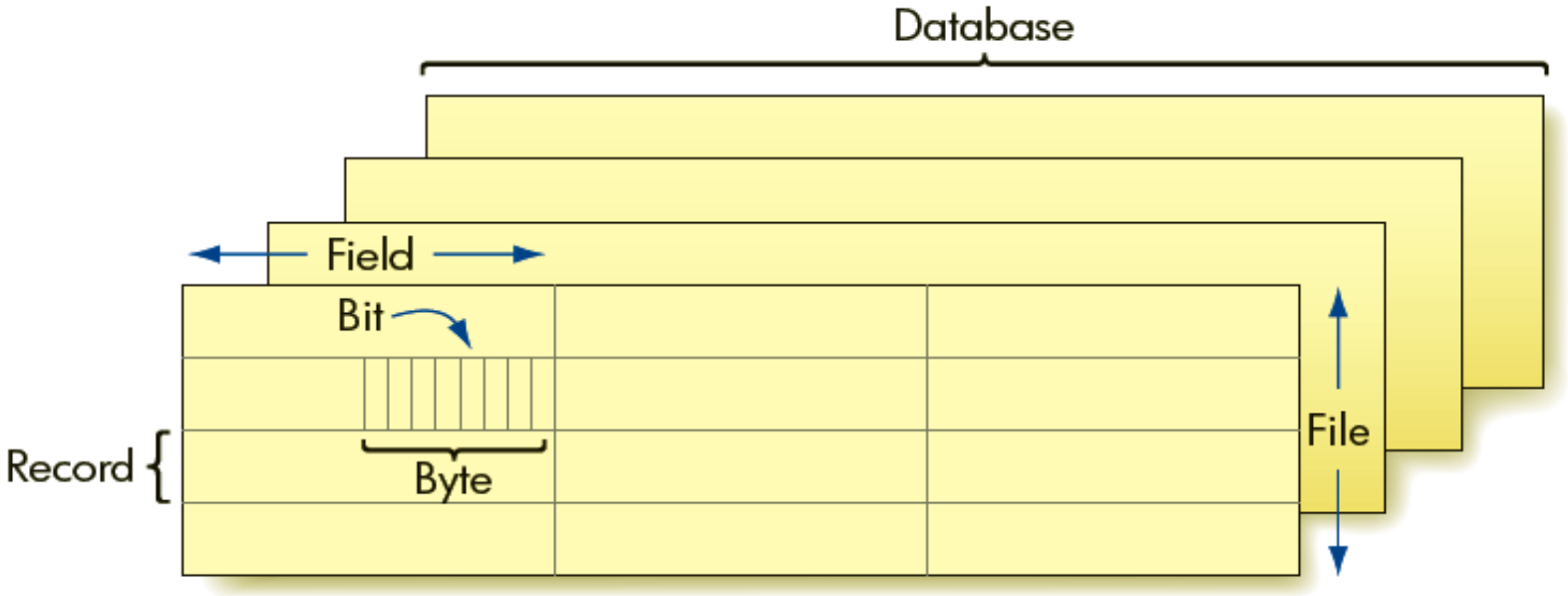


Figure 14.3 Data Organization Hierarchy

	Field 1	Field 2	Field 3
Record 1			
Record 2			
Record 3			
Record 4			
Record 5			

Figure 14.4 Records and Fields in a Single File

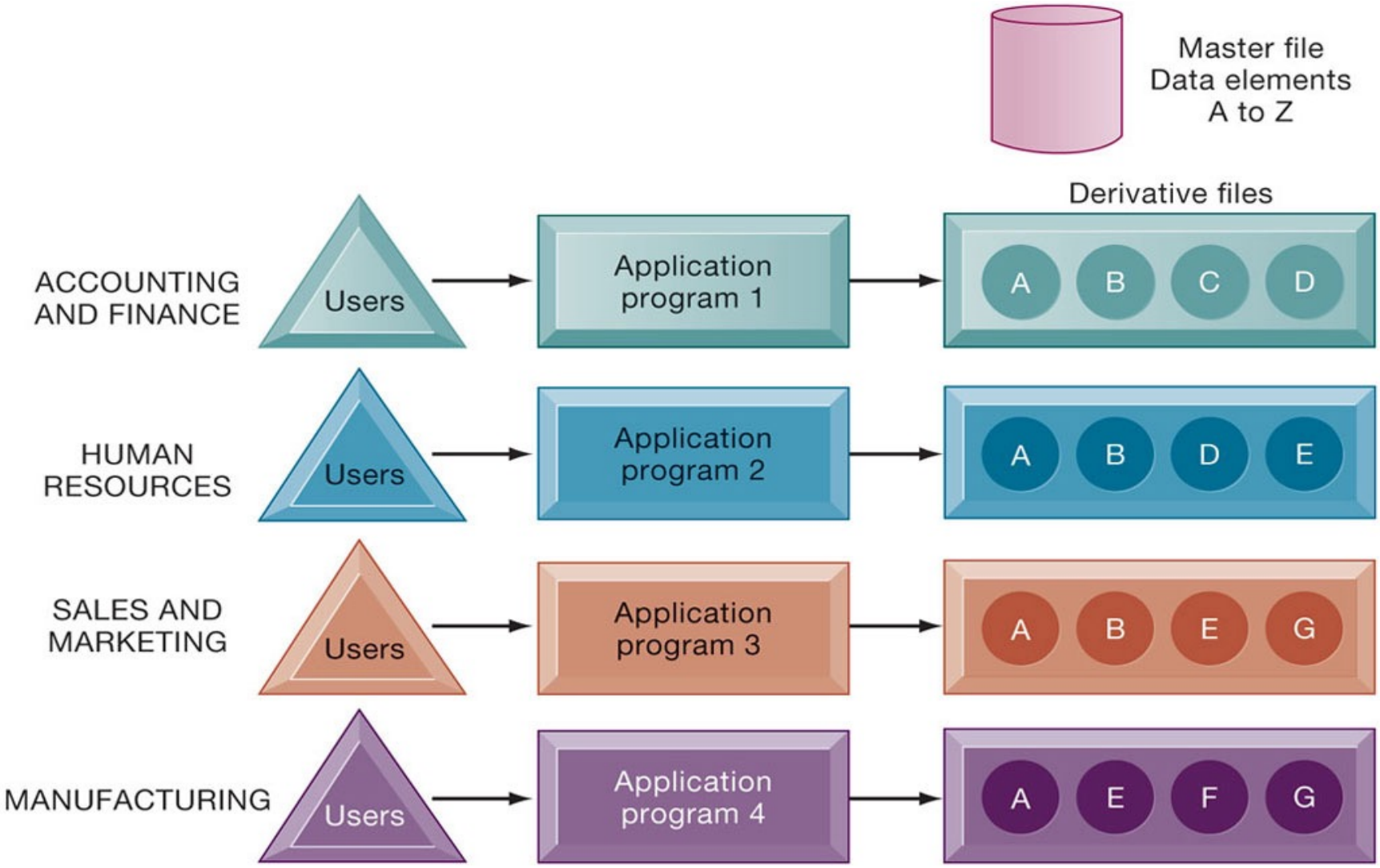
ID	LastName	FirstName	Birthdate	PayRate	HoursWorked
149	Takasano	Frederick	5/23/1966	$12.35	250

Figure 14.5 One Record in the Rugs-For-You Employees File

Problems with Traditional File Environments

- Files maintained separately by different owners
- Data redundancy
- Data inconsistency
- Program-data dependence
- Lack of flexibility
- Poor security
- Lack of data sharing and availability

Traditional File Processing



Database Management Systems

- Database
  - Serves many applications by centralizing data and controlling redundant data
- Database management system (DBMS)
  - Interfaces between applications and physical data files
  - Separates logical and physical views of data
  - Solves problems of traditional file environment
    - Controls redundancy
    - Eliminates inconsistency
    - Uncouples programs and data
    - Enables organization to centrally manage data and data security

Database Management Systems

- Manage the files in a database
- Entity
  - Fundamental distinguishable component
- Attribute
  - Category of information
- Primary key
  - Attribute or combination of attributes that uniquely identifies a **tuple**

EMPLOYEES

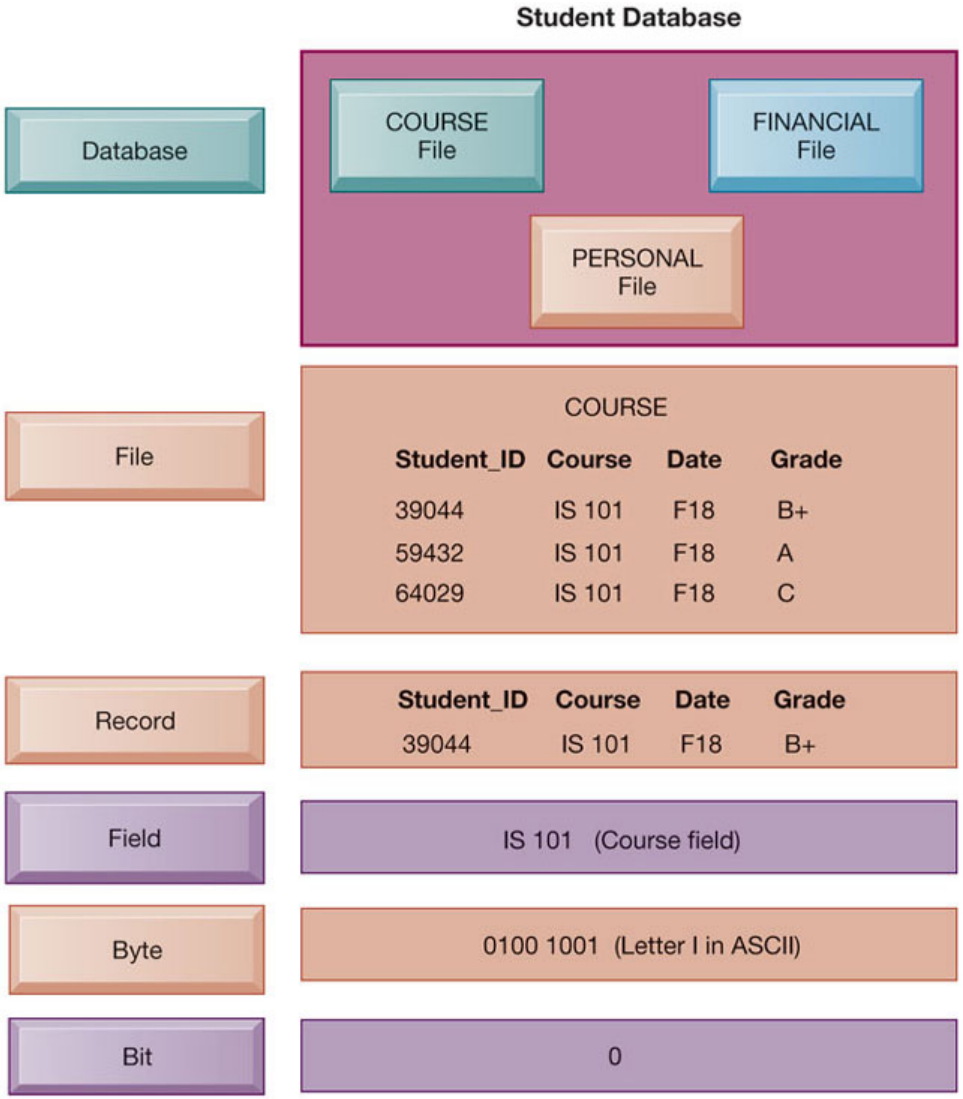
ID	LastName	FirstName	Birthdate	PayRate	HoursWorked
116	Kay	Janet	3/29/1956	$16.60	94
123	Perreira	Francine	8/15/1987	$ 8.50	185
149	Takasano	Frederick	5/23/1966	$12.35	250
171	Kay	John	11/17/1954	$17.80	245
165	Honou	Morris	6/9/1988	$ 6.70	53

Figure 14.6 Employees Table for Rugs-For-You

Terms and Concepts – Similar Definitions

- Database: Group of related files
- File: Group of records of same type
- Record: Group of related fields
- Field: Group of characters as word(s) or number(s)
- Entity: Person, place, thing on which we store information
- Attribute: Each characteristic, or quality, describing entity

The Data Hierarchy



Database Management Systems (cont'd)

- Query languages
  - Enable user or another application program to **query** the database, in order to retrieve information
- Composite primary key
  - Needed to identify a tuple uniquely
- Foreign key
  - Key from another table that refers to a specific key, usually the primary key

InsurancePolicies

EmployeeID	PlanType	DateIssued
171	B2	10/18/1974
171	C1	6/21/1982
149	B2	8/16/1990
149	A1	5/23/1995
149	C2	12/18/1999

Figure 14.7 Insurance Policies Table for Rugs-For-You

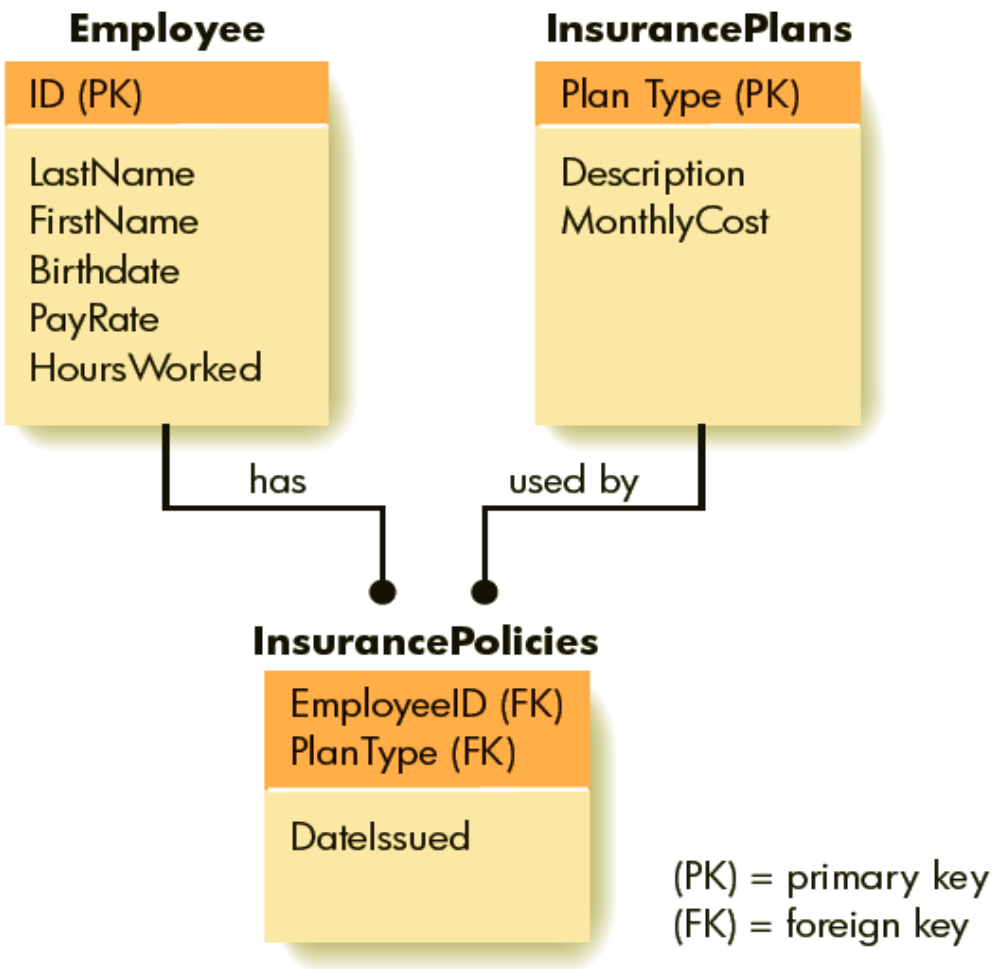


Figure 14.8 Three Entities in the Rugs-For-You Database

Functionality of a DBMS

The programmer sees SQL, which has two components:

- Data Definition Language - DDL
- Data Manipulation Language - DML
  - query language

Behind the scenes the DBMS has:

- Query engine
- Query optimizer
- Storage management
- Transaction Management (concurrency, recovery)

How the Programmer Sees the DBMS

- Tables:

Students:

SSN	Name	Category
123-45-6789	Charles	undergrad
234-56-7890	Dan	grad
	…	…

Takes:

SSN	CID
123-45-6789	CSE444
123-45-6789	CSE444
234-56-7890	CSE142
	…

Courses:

CID	Name	Quarter
CSE444	Databases	fall
CSE541	Operating systems	winter

"*data independence*": separate *logical* view from *physical implementation*

- Still implemented as files, but behind the scenes can be quite complex

Queries

- Find all courses that “Mary” takes
  - SELECT C.name
    FROM Students S, Takes T, Courses C
    WHERE S.name=“Mary” and
    S.ssn = T.ssn and T.cid = C.cid
- What happens behind the scene ?
  - Query processor figures out how to answer the query efficiently.

Transactions

- A *transaction* = sequence of statements that either all succeed, or all fail

- Transactions have the ACID properties:

  A = atomicity (a transaction should be done or undone completely)

  C = consistency (a transaction should transform a system from one consistent state to another consistent state)

  I = isolation (each transaction should happen independently of other transactions)

  D = durability (completed transactions should remain permanent)

Relational DBMS – Definitions

- Represent data as two-dimensional tables
- Each table contains data on entity and attributes
- Table: grid of columns and rows
  - Rows (tuples): Records for different entities
  - Fields (columns): Represents attribute for entity
  - Key field: Field used to uniquely identify each record
  - Primary key: Field in table used for key fields
  - Foreign key: Primary key used in second table as look-up field to identify records from original table

Relational Database Tables: Another Example

SUPPLIER

Columns (Attributes, Fields)

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_State	Supplier_Zip
8259	CBM Inc.	74 5th Avenue	Dayton	OH	45220
8261	B. R. Molds	1277 Gandolly Street	Cleveland	OH	49345
8263	Jackson Composites	8233 Micklin Street	Lexington	KY	56723
8444	Bryant Corporation	4315 Mill Drive	Rochester	NY	11344

Rows (Records, Tuples)

Key Field (Primary Key)

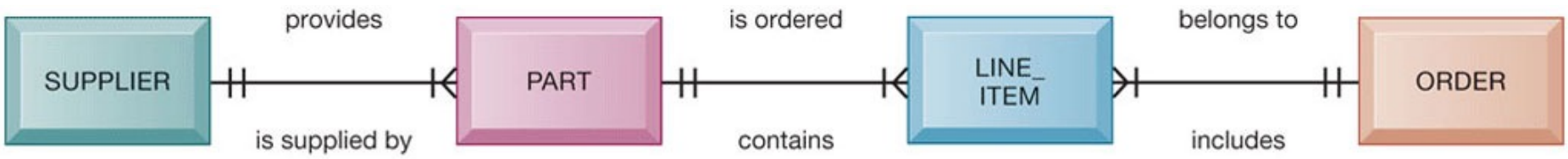
PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Primary Key

Foreign Key

Entity-Relationship Diagram: Another Example



SQL Query: Another Example

```
SELECT PART.Part_Number, PART.Part_Name, SUPPLIER.Supplier_Number,
SUPPLIER.Supplier_Name
FROM PART, SUPPLIER
WHERE PART.Supplier_Number = SUPPLIER.Supplier_Number AND
Part_Number = 137 OR Part_Number = 150;
```

Result of SQL Query: Another Example

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Select Part_Number = 137 or 150

SUPPLIER

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_State	Supplier_Zip
8259	CBM Inc.	74 5th Avenue	Dayton	OH	45220
8261	B. R. Molds	1277 Gandolly Street	Cleveland	OH	49345
8263	Jackson Components	8233 Micklin Street	Lexington	KY	56723
8444	Bryant Corporation	4315 Mill Drive	Rochester	NY	11344

Join by Supplier_Number

Part_Number	Part_Name	Supplier_Number	Supplier_Name
137	Door latch	8259	CBM Inc.
150	Door molding	8263	Jackson Components

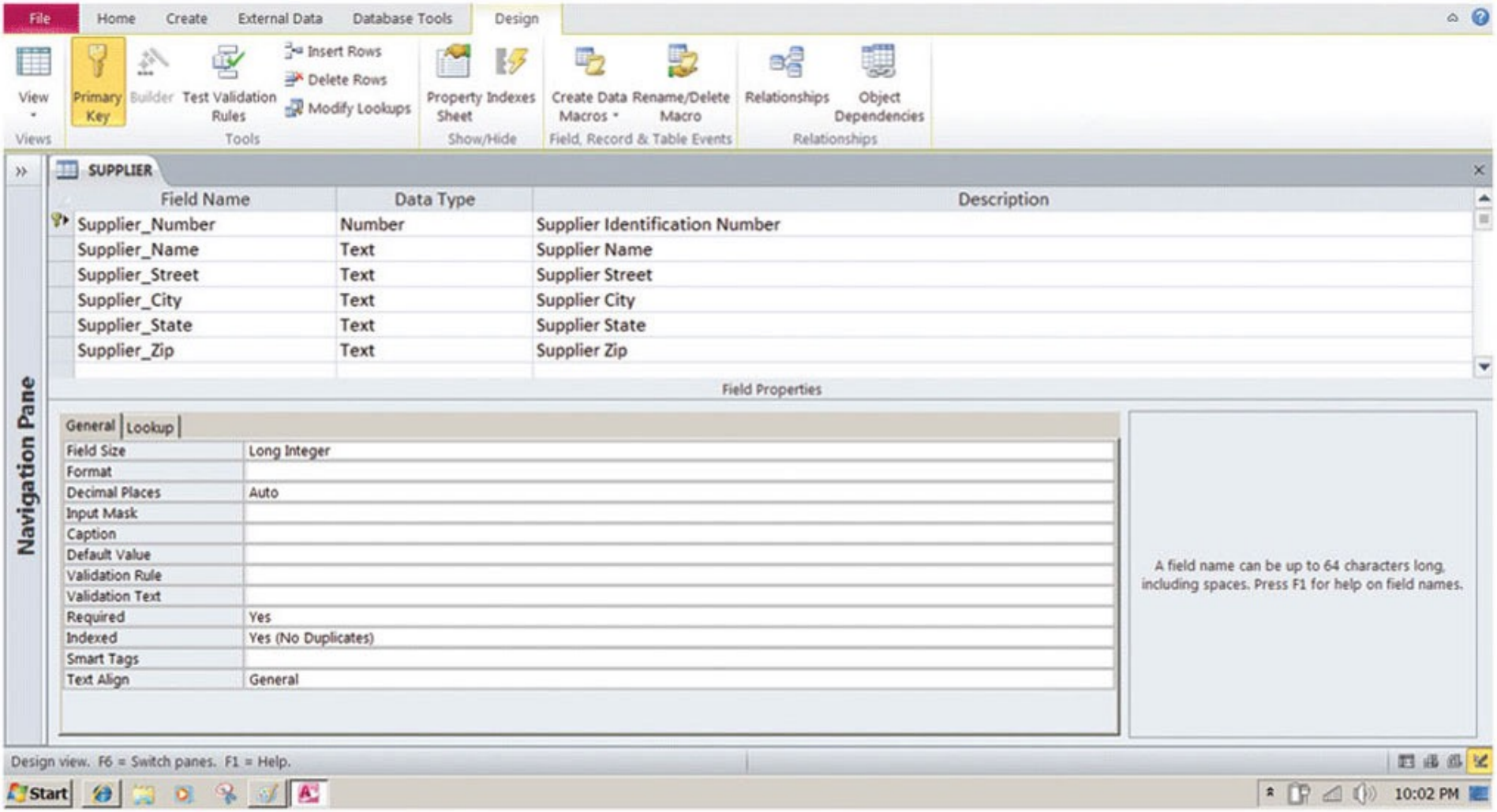
Project selected columns

Database Systems

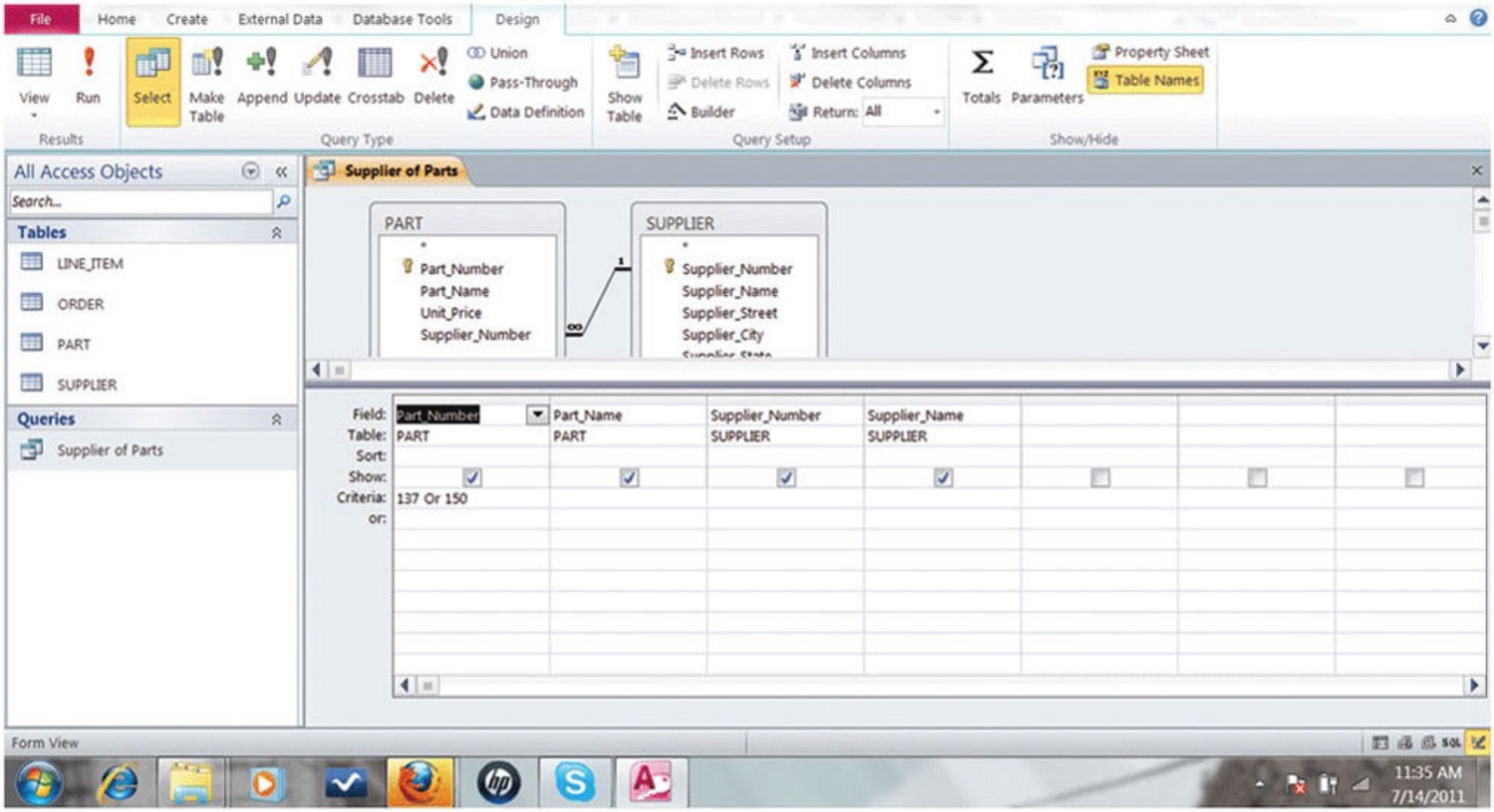
- The big commercial database vendors:
  - Oracle
  - IBM (with DB2)
  - Microsoft (SQL Server)
  - Sybase

- Some free database systems:
  - Postgres
  - MySQL
  - Predator

Example: Microsoft Access Data Dictionary Features



Example: Creating a Microsoft Access Query



Other Considerations

- Performance issues
  - Affect the user's satisfaction with a database management system
- To significantly reduce access time
  - Create additional records to be stored along with the file
- Distributed databases
  - Allow the physical data to reside at separate and independent locations that are electronically networked together

Non-Relational Databases and Databases in the Cloud

- Non-relational databases: “No SQL”
  - More flexible data model
  - Data sets stored across distributed machines
  - Easier to scale
  - Handle large volumes of unstructured and structured data
- Databases in the cloud
  - Appeal to start-ups, smaller businesses
  - Amazon Relational Database Service, MS SQL Azure
  - Private clouds

Summary

- Database
  - Allows data items to be stored, extracted, sorted, and manipulated
- Relational database model
  - Conceptual model of a file as a two-dimensional table